

Deep Learning approaches for finding a longest DG-consistent path in biological networks.

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Abstract. Systems biology is a developing scientific field that investigates living organisms as they naturally exist. Unlike traditional methods, it integrates information from multiple disciplines such as biology, physiology, and biochemistry to enhance the understanding of how organisms operate. This integration requires sophisticated and efficient methods for processing and interpreting data. A variety of techniques have been developed to compare biological networks, often utilizing representations derived from graph structures, in which vertices correspond to biological components and edges represent their interactions. This work addresses a computationally hard ($\mathcal{NP}$ -hard) problem involving heterogeneous biological networks, with a particular emphasis on how metabolic processes are influenced by genomic information. A directed graph D is used to represent the metabolic network, whereas an undirected graph G reflects gene proximity, both sharing the same vertex set. We propose deep learning-based methods, particularly leveraging graph neural networks, to identify paths in D such that the corresponding vertices are part of a connected subgraph in G . These paths represent sequences of metabolic reactions driven by enzymes encoded by genes located in close proximity, highlighting biologically meaningful patterns.

Introduction

To understand how living organisms function, researchers analyze the processes occurring within cells. Traditional approaches study different cellular aspects separately, such as proteomics, genomics, and genetics. Unlike traditional methods, systems biology adopts a holistic perspective by analyzing these components in an integrated manner. Its objective is to study biological entities holistically, considering them not merely as separate parts but as integrated systems exhibiting emergent behaviors. Research focus has moved away from isolated biological elements toward exploring the interactions among them. These interactions give rise to biological networks, with common examples including networks of protein interactions, neuronal networks, metabolic networks, and genetic interaction maps.

A biological network is represented as a graph, with vertices corresponding to biological components and edges or arcs indicating their interactions. The

variety of biological networks and the ways they are modeled have motivated the development of several comparison methods, many of which concentrate on homogeneous networks and make use of alignment techniques (see, for instance, Refs. [16,25,19,21]). These approaches have proven useful in identifying conserved modules across species-specific networks. Nevertheless, only a limited number of studies have addressed the comparison of heterogeneous networks [22,20,9,11,3,28,7,6,1].

We study the One-to-One SkewGraM problem, which arises as a key challenge in analyzing heterogeneous biological networks. The main objective is to uncover conserved metabolic and genomic signatures of a given species using two complementary biological networks, one modeled as a directed graph D and the other as an undirected graph G . The graph D typically encodes metabolic pathways, while G accounts for gene neighborhood or protein interactions [11,3,28]. Babou *et al.* [3] proposed a comparative framework that led to a specific combinatorial optimization problem. Given a directed graph $D = (V, A)$ and an undirected graph $G = (V, E)$, both defined over the same set of vertices $V = \{1, 2, \dots, n\}$, a path P in D is said to be (D, G) -consistent if the subgraph of G induced by the vertices of P is connected.

Several exact algorithms have been developed to address this problem, including branch-and-bound methods such as AlgoBB and AlgoBB++ [11,3,2,1], heuristic approaches like AlgoH [3], and integer linear programming models ILP1 and ILP2 [2]. These techniques vary in complexity and performance, with ILP2 often providing the most efficient and accurate results. Additionally, constraint programming methods have been proposed as alternative solutions [1], complementing these exact algorithms.

Deep learning has recently gained significant traction as a compelling alternative to traditional combinatorial optimization approaches. One significant development is the introduction of Pointer Networks [27], an end-to-end differentiable architecture aimed at tackling sequence-based combinatorial problems like the Travelling Salesman Problem (TSP). This line of research has been further extended through reinforcement learning techniques, exemplified by Neural Combinatorial Optimization [5], and neuro-symbolic approaches like NeuroSAT [23], which demonstrated the capacity of neural models to approximate solutions for NP-complete problems. More recently, Graph Neural Networks (GNNs) have gained traction in addressing graph-based combinatorial challenges. For instance, Dai et al. [8] developed a GNN framework to learn heuristics for the Minimum Vertex Cover problem, while Karalias and Loukas [15] proposed an unsupervised GNN-based method tailored to Max-Cut and other classical optimization problems on graphs.

Although significant progress has been made, to our knowledge, no deep learning method has yet been explicitly designed to address the *One-to-One SkewGraM* problem, which involves finding a path that simultaneously satisfies two constraints: it must be a valid directed path in a DAG D , and it must induce a connected subgraph in an undirected graph G . This problem does not naturally reduce to classical shortest or longest path problems and presents

additional complexity due to the necessity of ensuring consistency across two distinct graph structures. As such, this class of problems remains unexplored in the current deep learning and GNN literature, highlighting a significant gap that motivates our proposed approach.

In this paper, we present a new deep learning framework to solve the *One-to-One SkewGraM* problem. The best proposed method combines multiple components: Graph Convolutional Networks (GCNs) to capture the topological and functional features of both graphs D and G , and a sequence generation module to produce paths that fulfill the required consistency constraints.

The main contributions of this work are threefold. First, we introduce a unified deep learning architecture that integrates features from both metabolic and genomic networks. Second, we demonstrate the ability of this model to predict long DG-consistent paths on synthetic. Third, we provide a comprehensive comparison with state-of-the-art combinatorial algorithms, highlighting the strengths and limitations of the deep learning approach in this context.

The remainder of this paper is organized as follows. Section 2 provides a formal definition of the One-to-One SkewGraM problem. Section 3 introduces a Random Forest-based baseline for DG-consistent path prediction. In Section 4, we describe our proposed deep learning framework, covering data preprocessing, model architecture, and training procedures. Section 5 presents experimental results and compares our method with classical approaches. Finally, Section 6 concludes the paper and discusses directions for future research.

1 Problem Formulation: The One-to-One SkewGraM Problem

In this section, we introduce the combinatorial optimization problem called the *One-to-One SkewGraM problem*, initially proposed by Babou *et al.* [3], which has been shown to be $\mathcal{NP}$ -hard. The problem is formulated as follows.

Let:

- $D = (V, A)$ be a directed acyclic graph (DAG) representing directional biological processes, such as metabolic pathways;
- $G = (V, E)$ be an undirected graph encoding functional or physical interactions between identical biological entities, such as gene proximity.

The objective is to find a path $P = (v_1, v_2, \dots, v_k)$ in D that satisfies the following two conditions:

1. **Path condition:** $(v_i, v_{i+1}) \in A$ for all i , i.e., P is a valid directed path in D ;
2. **Connectivity condition:** The subgraph of G induced by the vertices of P is connected.

This formulation ensures that the resulting path respects both the causal flow described in D and the structural coherence represented in G .

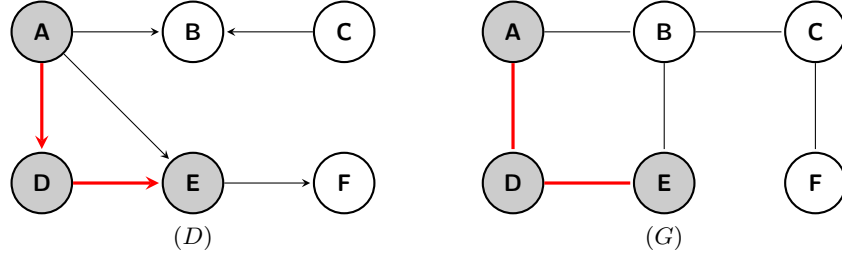


Fig. 1: Example. The longest path in D , $[A \rightarrow D \rightarrow E \rightarrow F]$, is not (D, G) -consistent since the induced subgraph on $\{A, D, E, F\} \subseteq V$ is not connected in G . The longest consistent path is $[A \rightarrow D \rightarrow E]$.

For example, consider the directed path $[A \rightarrow D \rightarrow E]$ shown in Figure 1.

Due to the dual constraints of respecting both directionality and topological coherence, the One-to-One SkewGraM problem is known to be $\mathcal{NP}$ -hard [12]. As a result, heuristic and learning-based approaches are required when dealing with large-scale biological networks.

2 Random Forest-Based Approach

In this approach, we aim to predict paths that are *DG-consistent*, i.e., paths that respect the topological ordering in a directed acyclic graph D and induce a connected subgraph in an undirected graph G .

Data Preparation. Each instance consists of a DAG D , an undirected graph G , and a reference path. For each node, we compute a set of topological and structural features:

- Degree in D and G
- Source/sink indicator in D
- PageRank score in G
- Degree centrality in G

These features are assembled into a feature matrix X and a label vector y , where $y_i = 1$ if node i belongs to the reference path and 0 otherwise.

Model Training. A Random Forest classifier is trained on the node-level features to predict whether a node belongs to the path. The model outputs binary predictions for each node.

Path Generation. To construct a DG-consistent path from node-level predictions:

1. Select all nodes predicted as part of the path.

2. Sort them according to a topological order of D .
3. Sequentially add nodes to the path, ensuring that the induced subgraph in G remains connected. If adding a node would break connectivity, it is skipped.

DG-Consistency Check. Finally, the predicted path is verified for DG-consistency:

- Topological order in D is respected.
- The subgraph induced by the path in G is connected.

3 Proposed Deep Learning Approaches

In this section, we present two approaches: the first requires post-processing, whereas the second directly predicts the DG-consistent path without any post-processing.

3.1 First approach

We propose **DGConsistentModel**, a neural architecture designed to predict biologically consistent paths across two complementary graphs representing distinct but related interaction modalities. These graphs are defined over the same set of biological entities (e.g., genes), and each captures a different aspect of biological relationships:

- A *directed graph* $D = (V, E_D)$, representing causal or hierarchical interactions (e.g., reaction flows in metabolic pathways).
- An *undirected graph* $G = (V, E_G)$, encoding physical or functional associations such as gene proximity or co-expression.

Each node $v \in V$ is represented by a feature vector combining:

- its degree in D ,
- its degree in G ,
- a binary indicator for being a source or a sink in D .

These features are concatenated into a 3-dimensional vector and serve as input to two parallel GCN-based encoders.

The architecture consists of two independent **Graph Convolutional Networks (GCNs)**:

- The *D-encoder* learns directional embeddings from the DAG D , capturing causal structure.
- The *G-encoder* processes the undirected graph G , focusing on neighborhood-based physical or functional connectivity.

Each encoder contains two GCN layers with ReLU activations. The resulting embeddings h_D and h_G are concatenated for each node, forming a joint representation. This representation is passed through a fully connected layer followed by a sigmoid activation, yielding a scalar score $\hat{y}_v \in [0, 1]$ for each node.

Training is performed with a binary cross-entropy loss, using ground-truth paths as supervision. At **inference** time, the predicted node scores are ranked and used to extract a path that satisfies two constraints:

1. The path must respect the topological order in D .
2. The nodes must form a connected subgraph in G .

By combining structural signals from both graphs, the DGConsistentModel learns to identify biologically plausible interaction chains that are consistent with both mechanistic and associative information [3,17,24].

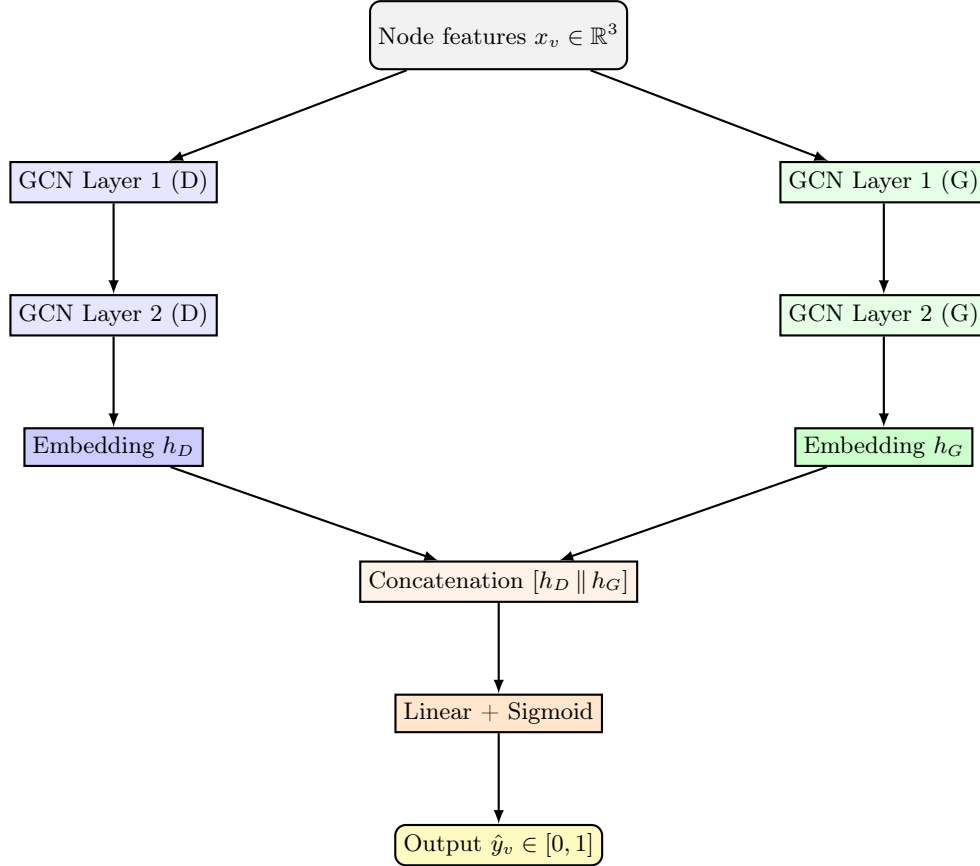


Fig. 2: DGConsistentModel architecture combining two GCN encoders for directed and undirected graphs.

3.2 Second approach

We propose a deep learning model to generate DG-consistent paths within biological networks by integrating structural and feature information from two related graphs. The architecture consists of a graph convolutional encoder and a sequential decoder with attention.

The model receives as input two graphs:

- A directed acyclic graph $D = (V_D, A_D)$ modeling causal biological processes.
- An undirected graph $G = (V_G, E_G)$ representing functional associations.

We consider the intersection of node sets $V = V_D \cap V_G$. For each node $v \in V$, a feature vector is computed including degree in D and G , source/sink indicator in D , PageRank, and degree centrality in G [18,13].

A two-layer Graph Convolutional Network (GCN) [18] is applied on D to embed node features into latent representations $H \in \mathbb{R}^{|V| \times h}$, capturing both topological and feature-based context.

A Long Short-Term Memory (LSTM) network [14] with an attention mechanism [4] sequentially generates a path:

- Initialized with a learned start token.
- At each step, attention scores candidates by combining decoder hidden state and candidate node embeddings.
- Candidate nodes are restricted to those consistent with topological order in D , connectivity in G , and unvisited status.
- An End-Of-Sequence (EOS) token allows terminating the path.

Attention mechanisms on graphs, such as Graph Attention Networks (GAT) [26], inspire the use of learned attention scores to prioritize candidate nodes in the decoding process.

The model is trained with teacher forcing using Negative Log-Likelihood loss over valid candidates at each decoding step. At inference, paths are generated greedily until EOS or maximum length [27].

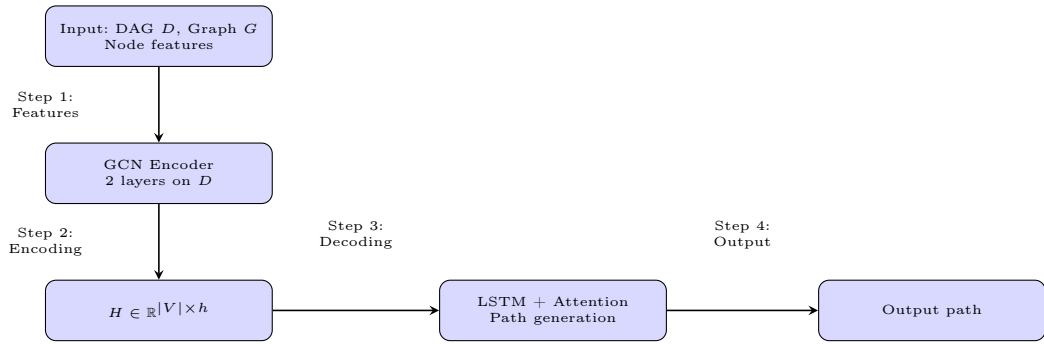


Fig. 3: Compact overview of DG-consistent path generation.

4 Experimental Results

In order to evaluate the performance of different methods for finding paths in graphs, we conducted experiments on a set of randomly selected instances.

The methods compared include ILP2 developed in [2], which serves as the ground truth by providing the exact path length, a Random Forest model (RF), and two deep learning approaches (DL1 and DL2). The table below summarizes the path lengths obtained by each method for 10 randomly selected instances, allowing for a direct comparison of their performance.

To evaluate the proposed GNN model for predicting DG-consistent paths, we conducted extensive experiments on synthetic graph datasets generated using classical random graph models: Erdős-Rényi (ER).

Dataset generation. We generated 1000 instances, each consisting of a pair of graphs (D, G) sharing the same set of $N = 50$ nodes. The DAG D was constructed as follows:

- **Erdős-Rényi graphs:** We generated a random directed acyclic graph D by first sampling a random topological ordering of nodes, then adding edges between nodes with probability $p = 0.05$ only from earlier to later nodes to ensure acyclicity [10].

Experiments were conducted on a dataset composed of graph instances tailored for the DG-consistent path prediction task. Each instance includes a directed acyclic graph D , an undirected graph G sharing a subset of nodes with D , and a ground truth solution path consistent with both graphs.

The dataset contains 1000 instances, split into 800 training instances indexed from 50_1 to 50_800 and 200 test instances indexed from 50_801 to 50_1000. Each graph comprises approximately 50 nodes modeling biological network interactions.

We employed the first approach, a dual-graph graph convolutional network that processes D and G separately through two GCN layers each. Node embeddings from both graphs are concatenated and passed through a linear layer with sigmoid activation to predict node membership probabilities for the DG-consistent path.

Node features include degree values in both D and G , as well as a binary source/sink indicator in D .

The model was trained for 300 epochs using binary cross-entropy loss with the Adam optimizer (learning rate 0.01). Each training batch corresponds to a single graph instance with node-level labels derived from the ground truth path.

Predicted node scores are converted into DG-consistent paths using a heuristic search starting from the top hub nodes (those with highest combined degree in D and G). The search iteratively adds nodes respecting the following constraints:

- Existence of an edge from the current last node to the candidate node in D .

- Connectivity preservation of the induced subgraph on G .

This approach ensures paths comply with both the partial order on D and connectivity requirements on G .

The second approach was evaluated on the same benchmark datasets and experimental settings as the previous model. The implementation involves loading the directed acyclic graph D and the undirected graph G , extracting node features such as degree, centrality measures, source/sink indicators, and PageRank, which are encoded by a two-layer Graph Convolutional Network (GCN). A sequential decoder based on an LSTM with attention mechanism is trained to generate DG-consistent paths by selecting candidate nodes that respect topological order and connectivity constraints. The training uses a cross-entropy loss with teacher forcing over ground-truth sequences.

Although the model successfully generates DG-consistent paths, the predicted paths tend to be relatively short compared to the reference solutions. This limitation may arise from conservative candidate selection during decoding or insufficient exploration during training, often resulting in truncated path predictions. These results suggest that while the model captures key structural constraints, further improvements in candidate selection strategies, training procedures, or architectural design are needed to promote the generation of longer, biologically relevant paths.

Table 1 summarizes the comparison of methods on 10 randomly selected instances. Each instance represents a directed graph D and an associated undirected graph G . The number of arcs in D and edges in G are denoted by $|A|$ and $|E|$, respectively.

The methods compared include ILP2, which serves as the ground truth by providing the exact path length, a Random Forest model (RF), and two deep learning approaches (DL1 and DL2). The table below summarizes the path lengths obtained by each method.

Instance	$ A $	$ E $	$ Path _{ILP2}$	$ Path _{RF}$	$ Path _{DL1}$	$ Path _{DL2}$
1	100	106	3	0	2	0
2	91	111	4	0	2	0
3	96	97	5	0	5	2
4	93	97	2	0	2	0
5	110	93	2	0	2	0
6	104	81	6	0	2	2
7	95	106	2	0	0	0
8	118	103	3	0	3	0
9	113	104	4	0	2	0
10	103	104	6	0	3	0

Table 1: Comparison of methods on 10 test instances: $|A|$ arcs in D , $|E|$ edges in G , $|Path|$ path length found by ILP2 (ground truth), RF, DL1 and DL2.

As shown in Table 1, ILP2 consistently provides the exact path length for each instance, serving as the reference. Random Forest (RF) fails in all cases (all zeros), indicating its ineffectiveness. Deep Learning 1 (DL1) partially predicts paths, sometimes matching the ground truth (e.g., instance 3), while Deep Learning 2 (DL2) correctly predicts only DG-consistent paths (e.g., instances 3 and 6), otherwise returning 0. Overall, ILP2 is the most reliable method, whereas DL1 and DL2 show limited potential for approximating paths.

5 Conclusion

We proposed a novel GNN-based model to address the One-to-One SkewGram problem, effectively integrating both directed and undirected structures of biological graphs. The results demonstrate that both studied approaches ensure perfect DG-consistency, confirming that the predicted paths induce connected subgraphs in G . However, they differ significantly in average predicted path length: the first approach generates longer paths (7.4 nodes on average) requiring post-processing, while the second directly predicts shorter DG-consistent paths (1.2 nodes on average) without additional steps. This trade-off between path length and computational simplicity highlights the need for further biological validation to determine the practical utility of shorter versus longer predicted paths.

Future Work

- Scaling the methods to larger and dynamic biological graphs.
- Incorporating advanced attention mechanisms to improve interpretability of predictions.
- Extending the framework to handle multi-relational or heterogeneous biological networks.
- Exploring hybrid approaches combining the strengths of both path length and direct prediction.

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